University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing*

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester I— 2022

SCS1206 — Laboratory I

(Two (2) Hours)

Answer ALL questions

Number of Pages = 14 Number of Questions = 4

To be completed by the candidate	
Index Number	

22?

Important Instructions to candidates:

- The medium of instruction and questions is **English**. Students should answer in the medium of English language only.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number on each and every page of the question paper.
- The duration of the paper is **Two (2)** hours.
- The paper has **4** questions on **14** pages.
- Answer **all** the questions.
- **Write your answers on the space provided** on this question paper.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.

To be completed by the examiners

1	
2	
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1. (a). i. Draw a diagram of the layers of a computer system indicating the hardware, the operating system and the software applications including any sub-categories of these (either as a stack or as concentric circles) and name each such layer.

[3 marks]

ii. What is referred to as the kernel of an operating system?

[2 marks]

(b). Give three (03) examples of each of the software referred to in following components in the Windows and Linux ecosystems.

i. Tools and utilities.

[2 marks]

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ii. Application software.

[2 marks]

(c). Briefly explain what you understand by the following, giving examples to illustrate your answer.

i. Desktop environment.

[2 marks]

ii. Shell.

[2 marks]

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iii. Linux distribution.

[2 marks]

(d). i. List 3 different ways in which you can install and run a Linux system on a computer which currently only has Windows.

[3 marks]

ii. In which way(s) is installing applications on Linux different from installing software on Windows?

[3 marks]

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iii. Explain two (02) different ways in which software can be installed on Linux.

[4 marks]

2. (a). What do the following BASH commands do?

i. `ls -al`

[1 marks]

ii. `cat file1.txt file2.txt`

[1 marks]

iii. `mv file1.txt file2.txt`

[1 marks]

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iv. `mv file1.txt file2.txt file`

[1 marks]

v. `head -n12 file.txt`

[1 marks]

vi. `wc file.txt`

[1 marks]

vii. `mkdir docs pics; cd pics`

[1 marks]

viii. `rm -r docs`

[1 marks]

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ix. `touch file1.txt file2.txt file3.txt`

[1 marks]

x. `cat > file.txt`

[1 marks]

(b). Give BASH commands needed to achieve the following tasks.

i. Display the username of whoever is logged into the system.

[2 marks]

ii. To change the permission of a script to be executable by you, your group and the world, but only readable and editable by you.

[2 marks]

iii. To change the current directory to the subdirectory called `pics` in your home directory.

[2 marks]

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iv. To re-execute the command number `1324` in your history.

[2 marks]

v. To execute the command `cat` on the last argument of whatever was the last command executed in the shell.

[2 marks]

vi. To re-execute the last command again.

[2 marks]

vii. To display the edits made on a file called `text1.txt` which resulted in the file `file2.txt`.

[2 marks]

viii. To display the files in the current directory in reverse alphabetical order.

[2 marks]

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ix. To display the files in the current directory in ascending order of the file size.

[2 marks]

x. To display the files in the current directory with the most recent files first down to the oldest files last.

[2 marks]

3. (a). Assume that you are given a book on Linux as a text file `linux.txt`.

i. Give a command needed to find the number of words in the file.

[2 marks]

ii. Give a command to list all lines in the file that end with a period (full stop).

[2 marks]

iii. Give a command to list all lines in the file that do NOT contain the words `compute`, `computing`, `computation`, and `computer`.

[2 marks]

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iv. Give a (piped) command to convert the file into a list of distinct words sorted in descending order of their frequency.

[3 marks]

v. Give a (piped) command to convert the file into a list of distinct words sorted by rhyming order.

[3 marks]

(b). Briefly explain the following BASH features giving examples.

i. || and && commands.

[2 marks]

ii. The `.bashrc` file.

[3 marks]

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iii. The `alias` command.

[3 marks]

iv. `CTRL-C`, `CTRL-D` and `CTRL-Z` commands.

[3 marks]

v. `bg` and `fg` commands.

[2 marks]

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4. Give BASH scripts to achieve the following tasks.

(a). The script `hello` should greet the user by their username and list the files in their home directory with a suitable message.

[3 marks]

(b). The script `whatsin` that takes a text file as an argument and gives the number of words in it. For example, executing the script as `./whatsin linux.txt` should display `The file linux.txt has 12471 words` (assuming that the file `linux.txt` has a total of 12,471n words in it).

[4 marks]

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(c). A script `add` that takes in two arguments and displays their sum with a suitable message. For example, executing the script as `./add 23 65` should display the message `The sum of 23 and 65 is: 88.`

[5 marks]

(d). The script `files` should check first to see if the file given to it as an argument exists in the current directory and if so, check if the user has permission to execute it. For example, calling it as `./files myfirst` should display the message `The file myfirst exists` and then `It is executable` or `It is not executable` as the case may be.

[5 marks]

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(e). The script `args` should display the number of arguments passed to it and then list them out. For example, `./args treat hallogen 23 4T3` should display the text `There are 4 arguments` followed by the text `They are treat, hallogen, 23, 4T3.`

[3 marks]

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